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Heterogeneous Multi-Task Learning for Real-Time Collaborative Prediction of Multiple Interrelated Drilling Parameters

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Abstract

Accurate real-time prediction of drilling parameters, such as Rate of Penetration (ROP), Bottom Hole Pressure (BHP), and Formation Pressure (FP), is vital for optimizing drilling efficiency and ensuring operational safety. However, these parameters are interrelated, and traditional machine learning models often overlook their vital interdependencies, limiting predictive performance. We propose a novel Heterogeneous Multi-Task Learning (HMTL) framework to capture these complex non-linear interdependencies and enable synergistic co-prediction of these crucial drilling parameters. The HMTL framework introduces a novel architecture that intelligently integrates heterogeneous individual predictive models for ROP, BHP, and FP, each grounded in mechanism-data fusion. This architecture comprises: (1) task-specific expert networks, leveraging the strengths of diverse base models tailored to each parameter's unique characteristics; (2) a shared expert layer, distilling common underlying features from comprehensive drilling data; and (3) adaptive gating mechanisms, dynamically balancing task-specific insights with shared knowledge. Moreover, four loss balancing strategies were evaluated with hyperparameter tuning using Bayesian Optimization. Validation on field drilling datasets from the Tarim Oilfield demonstrates the HMTL model's superior performance. The Adaptive Weighting (AW) loss balancing strategy yielded the best results, achieving a Mean Absolute Percentage Error (MAPE) of 3.74% for ROP, 0.36% for BHP, and 1.45% for FP, with corresponding R^2 values of 0.985, 0.976, and 0.929, respectively. Notably, the HMTL approach significantly outperformed the standalone Single-Task Learning (STL) models, reducing the prediction error for ROP by approximately 50%, and for BHP and FP by approximately 30% and 28%, respectively. This enhanced accuracy was achieved while also reducing the overall model training time by approximately 28%, underscoring the efficiency and effectiveness gains of joint parameter prediction. This paper introduces a novel HMTL architecture that leverages interdependencies among multiple drilling parameters for enhanced collaborative prediction. It uniquely addresses the challenge of integrating diverse, specialized single-parameter models into a cohesive, efficient predictive system. This offers a transformative

framework for holistic, accurate, and computationally efficient intelligent drilling operations, scalable for future parameter additions.

Introduction

Optimizing the drilling process in complex environments, such as deep, ultra-deep, and unconventional reservoirs, is a key challenge in modern oil and gas exploration. Intelligent drilling, which integrates artificial intelligence with petroleum engineering, offers a transformative approach to address these challenges (Li et al., 2022). A primary application involves using intelligent models for precise characterization, real-time prediction, and optimized decision-making for crucial physical parameters during drilling (Abughaban et al., 2019; Ajimoko, 2025)

Drilling is a complex, dynamic process involving the coupling of multiple physical fields, encompassing drill bit-rock interaction, downhole drill string dynamics, multiphase fluid mechanics in the wellbore, and formation response (as shown in Fig. 1) (Li et al., 2022). Among these parameters, Rate of Penetration (ROP) (Ma et al., 2025), Bottom Hole Pressure (BHP) (Zhang et al., 2023) and Formation Pressure (FP) (Ahmed et al., 2019) are three key indicators for characterizing the drilling status. ROP directly reflects drilling efficiency, while BHP must be strictly controlled within the safe pressure window defined by FP and fracture pressure to prevent complications such as kicks, fluid loss, and wellbore instability. Therefore, achieving high-precision, real-time prediction of these three parameters is a technical prerequisite for ensuring safe and efficient drilling operations.

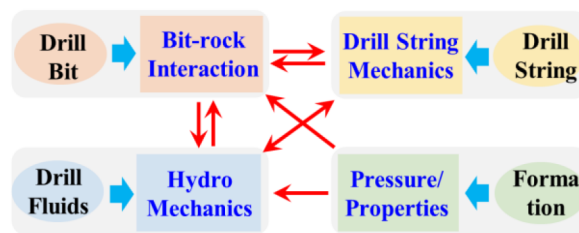


Figure 1—Primary subsystems and their interactions in the drilling process.

However, current AI-based research on drilling parameter prediction faces two primary limitations. First, the prevailing approach employs a Single-Task Learning (STL) paradigm, where a dedicated model is independently built and trained for each parameter, such as ROP. While this method has achieved considerable accuracy on individual tasks, it fundamentally treats an inherently coupled system as a collection of isolated problems. For instance, changes in ROP (rate of cuttings generation) directly affect the properties of the drilling fluid in the wellbore, which in turn alters the BHP. The STL approach overlooks this shared, underlying information, potentially limiting the predictive performance of the models and failing to provide a holistic understanding of the drilling process. Second, in practical applications, engineers must train, deploy, and manage multiple independent models to achieve comprehensive monitoring. This significantly increases both computational overhead and model management complexity.

While researchers have explored mechanism-based coupling of multiple drilling systems to achieve global optimization (Cayeux et al., 2021; Gandikota et al., 2023; Khazali-Rosli et al., 2024; Losoya et al., 2020; Saini, 2020), the effective collaboration of multiple heterogeneous data-driven models remains a largely unexplored area. Furthermore, modern intelligent models are not simple, single-architecture structures. They often deeply integrate physical mechanisms and domain knowledge, resulting in customized network architectures and loss functions, which makes their collaboration even more challenging (Yao et al., 2023; Zhang et al., 2024).

To address these challenges, this paper aims to fill this research gap by proposing a novel Heterogeneous Multi-Task Learning (HMTL) framework, based on the principles of Multi-Task Learning (MTL) (Caruana,

1997; Zhang & Yang, 2021). This framework treats the prediction of ROP, BHP, and FP as three highly related yet distinct, or heterogeneous, tasks, learning them collaboratively within a single, unified model. By sharing representations across tasks, the HMTL framework learns more generalizable underlying features. Its core innovation lies in its specific design to integrate base models that differ in their network structures, input features, and even training paradigms due to their distinct underlying physical mechanisms. Through a carefully designed architecture of task-specific networks, a shared knowledge layer, and an adaptive gating mechanism, our framework leverages the intrinsic interdependencies among tasks. The goal is to simultaneously improve predictive accuracy and significantly reduce computational costs, offering a more efficient and precise holistic solution for intelligent drilling.

Foundation: Heterogeneous Single-Task Models

The foundation of the HMTL framework consists of three individual intelligent models designed for ROP, BHP, and FP, each integrating mechanistic principles with data. The differences in their structures and principles constitute the "heterogeneity" challenge that multi-task learning must address. Each model's architecture is not arbitrary; it is intentionally designed to reflect the dominant physical principles governing the target parameter, a crucial aspect for building robust and interpretable models.

Mechanism-Data Fusion ROP Prediction Model

The ROP prediction model employs a sub-network structure inspired by the mechanism of rock breaking (Zhang et al., 2022). The core idea is to decouple the complex influencing factors into four main categories: formation, hydraulics, engineering parameters, and bit wear, and construct corresponding sub-networks for each (Fig. 2). The model consists of four expert sub-networks: HydroNet (a Fully Connected Neural Network, FCNN) captures the influence of hydraulic factors; GeoNet (an FCNN) characterizes formation lithology features; WearNet (an Attention-based Gated Recurrent Unit, GRU) models the dynamic sequence of drill bit wear. Finally, ROPNet (an FCNN) fuses the outputs of these sub-networks with key engineering parameters, such as weight on bit and rotational speed, to accurately predict ROP. This "divide-and-conquer" approach mirrors how a domain expert would analyze the problem, improving model interpretability and allowing each sub-network to specialize in learning a specific aspect of the complex bit-rock interaction. For example, the use of an Attention-based GRU for WearNet is deliberate; bit wear is a cumulative, time-dependent process where recent drilling conditions may have a greater impact than distant past conditions. The GRU captures this sequential dependency, while the attention mechanism allows the model to dynamically focus on the most influential periods of drilling when assessing the current state of wear. This tailored architecture allows the neural network to deeply mine the complex non-linear relationships between these factors and ROP, offering a more powerful representation capability than traditional mechanistic models. For more details, please refer to Zhang et al. (2022).



The BHP prediction model is a Physics-Informed Hybrid Neural Network (PIHNN) (Zhang, 2024). Its innovation lies in integrating fluid dynamics governing equations as a soft constraint into the model's training process (Fig. 3). The model is composed of an FCNN for processing spatio-temporal inputs and a GRU network for handling sequential relationships. Its loss function includes not only a data-driven term but also a physics-informed regularization term derived from the partial differential equations (PDEs) of solid-liquid two-phase flow, as well as empirical constraints from drilling knowledge. This design ensures that the model's predictions adhere to fundamental physical principles, such as mass and momentum conservation. The physics-based regularization term acts as a powerful constraint, preventing the model from making physically implausible predictions, especially in regions of the input space that are sparsely represented in the training data. This significantly enhances the model's stability, generalization ability, and trustworthiness, which are critical for safety-related parameters like BHP.



The design of the FP prediction model integrates domain knowledge from geological sedimentology, fully considering the strong spatial auto-correlation that formation pressure exhibits due to depositional patterns (Fig. 4) (Zhang, 2024). The model uses a hybrid architecture: an FCNN processes logging data that directly reflects formation lithology, while another GRU network specifically captures the sequential dependencies (i.e., auto-regressive relationships) in historical FP data. Finally, a top-level FCNN fuses the outputs from both to make the final prediction. The logic behind this hybrid structure is to capture two distinct sources of information. The FCNN branch processes "point-wise" data—such as gamma ray and resistivity logs—that indicate the rock type at a specific depth. Simultaneously, the GRU branch models the pressure trend as a sequence with respect to depth, acknowledging that the pressure at depth D is highly dependent on the pressure at $D-\Delta D$ due to geological compaction trends. This structure effectively combines the direct

influence of instantaneous formation properties with the inherent, depth-dependent trends in the pressure sequence.

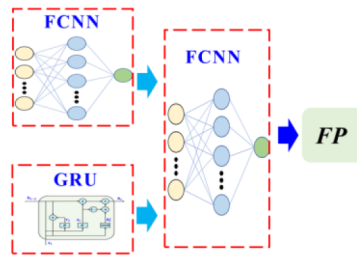


Figure 4—Architecture of the knowledge-based hybrid neural network for FP prediction.

Summary of Model Heterogeneity

The three foundational single-task models described above are fundamentally heterogeneous. This diversity is not a weakness but a strength, as each model is expertly tailored to its specific problem. The heterogeneity manifests in three key areas: (1) Architectural Paradigms: The models employ distinct architectures, from the modular sub-network design of the ROP model to the physics-constrained hybrid network for BHP and the auto-regressive sequence model for FP. (2) Input Feature Spaces: Each model is fed a customized set of input features carefully selected based on domain knowledge to be most relevant for predicting its target variable. (3) Learning Objectives: The training paradigms differ, most notably with the BHP model incorporating a unique physics-based penalty in its loss function, altering the optimization landscape.

The Heterogeneous Multi-Task Learning (HMTL) Framework

The models' inherent heterogeneity makes traditional MTL architectures, which often assume identical model structures, difficult to apply directly and is the primary motivation for designing the customized HMTL framework in this paper. To integrate these heterogeneous models and leverage their intrinsic correlations, we have designed a novel HMTL framework aimed at preserving the expertise of each model while enabling efficient knowledge sharing across tasks.

Multi-Task Learning and Its Applicability

The core idea of Multi-Task Learning (MTL) is to train a unified model to solve multiple related tasks simultaneously (Fig. 5). Compared to independent STL models, MTL offers significant advantages (Fei & Huan, 2013; Zhang & Yang, 2021): (1) It reduces model complexity and the number of parameters, which improves both computational and deployment efficiency. (2) Through a shared mechanism, MTL learns a common representation by extracting generic features beneficial to all tasks. This allows the model to learn a broader feature set, improve its performance on unseen data, mitigate overfitting, and significantly enhance its generalization ability.

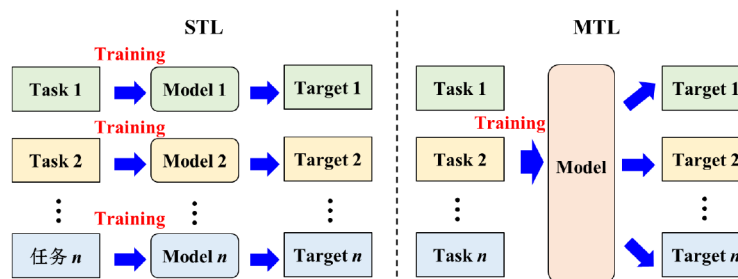


Figure 5—Single-Task Learning (STL) vs. Multi-Task Learning (MTL).

The core principle of Multi-Task Learning (MTL) is that tasks are inherently related. This means that while individual tasks can be addressed independently, they often represent different facets of a single, integrated system and therefore share underlying connections. The rationale and necessity of applying the MTL paradigm to the joint prediction of ROP, BHP, and FP are grounded in the following three key aspects: (1) System-level Correlation: These three prediction tasks represent different facets of a complete physical system: rock breaking, wellbore hydraulics, and formation response, respectively. (2) Physical Coupling Between Tasks: The subsystems have strong direct or indirect coupling relationships. For example, an increase in ROP leads to a higher concentration of drilled cuttings in the annulus, which increases the equivalent circulating density of the drilling fluid and, consequently, raises the BHP. Similarly, FP dictates the safe operating window for both BHP (to avoid kicks) and the weight-on-bit applied (which influences ROP), creating a tight operational coupling. (3) Common Data Source: Although the input parameters for each model differ, all data originate from the same well's Measurement-While-Drilling (MWD) and Logging-While-Drilling (LWD) systems, reflecting a unified downhole environment state. These factors satisfy the core assumption of MTL: that the tasks are related and can benefit from shared learning. Therefore, MTL is a suitable approach for the collaborative modeling of intelligent prediction models for ROP, BHP, and FP.

HMTL Framework Architecture

Given the heterogeneity of the base models, traditional hard-parameter sharing (where layers are directly shared) or soft-parameter sharing (where parameters are linked by regularization) MTL architectures are not directly applicable. Hard sharing is too restrictive for models with different architectures, while soft sharing may not be powerful enough to enforce strong knowledge transfer. To overcome this, we drew inspiration from modern Mixture-of-Experts (MoE) architectures. Our analysis (Fig. 6) reveals that all three models can be abstracted into two parts: a lower-level feature extraction part and an upper-level task prediction part. Based on this observation, we designed the customized HMTL architecture shown in Fig. 7. This architecture preserves the core structure of each original model while innovatively introducing a shared layer and gating units to enable the organic fusion of task-specific and shared information.

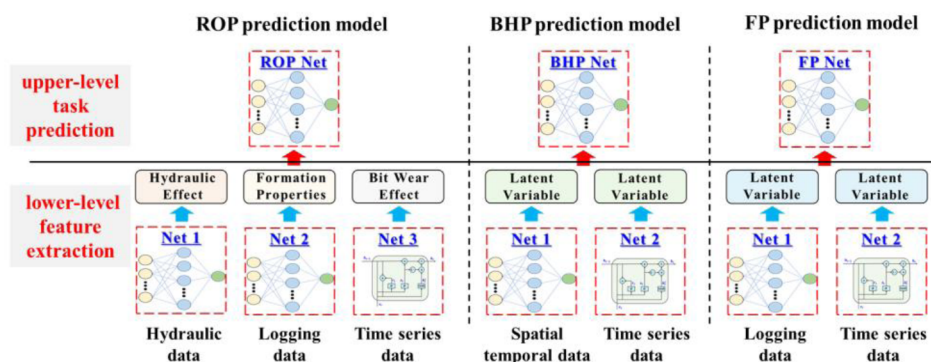


Figure 6—The heterogeneous intelligent models for ROP, BHP and FP prediction.

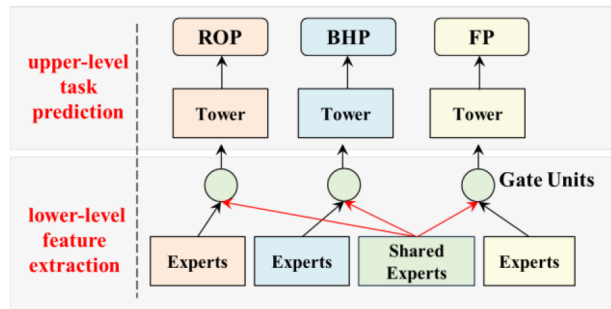


Figure 7—The proposed HMTL framework for collaborative modeling of multiple drilling processes.

Task-Specific Expert Networks and Tower Networks. Following the terminology of MTL, the original single-task models are divided into two parts: (1) Expert Networks (Experts): The lower-level feature extraction sub-networks of each model are repurposed as "task-specific experts." They are used to extract features most relevant to their specific task. As shown in Fig. 8, the initial FCNN and GRU in the BHP model serve as experts, while HydroNet, GeoNet, and WearNet act as experts for the ROP model. (2) Tower Networks (Towers): The upper-level prediction sub-networks are referred to as "towers." They are responsible for integrating the information extracted by the lower-level networks to predict the target parameter. For example, ROPNet is the tower network for the ROP task.

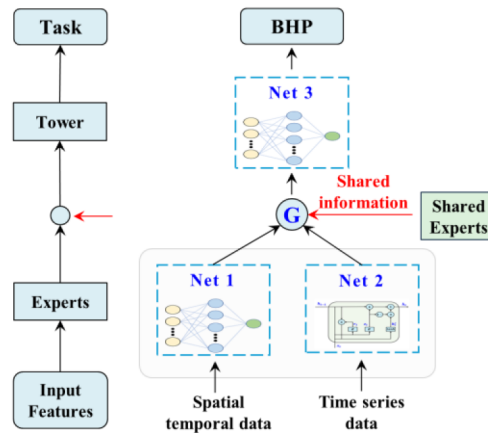


Figure 8—Expert and Tower networks within the HMTL framework.

Shared Layer and Gating Mechanism. To enable knowledge sharing, we introduced two key components: (1) Shared Layer: This layer aims to extract generic features beneficial to all tasks. It consists of multiple attention-based GRU networks that function as "shared experts." Its input is the union of all input parameters from the three tasks, allowing it to learn a comprehensive representation of the overall drilling state. Its output is a state vector representing shared, underlying information. The choice of attention-based GRUs here is crucial, as it allows this shared layer to act as a holistic "system state encoder," learning to dynamically weigh the importance of different parameters over time to form a rich, contextual understanding of the entire drilling process. (2) Gating Mechanism: Each task is equipped with a dedicated gating unit. This unit is a simple FCNN with a SoftMax output that dynamically controls the flow of information. As shown in Fig. 9, it takes the outputs from both the task-specific expert network O_k and the shared layer O_s as input, then generates two weight vectors (w_1^{gk} and w_2^{gk}). These weights are used to compute a weighted sum of the task-specific and shared information, as formulated in Equation (1). This gating mechanism functions as a learned "information arbiter." It allows the model to decide, for each task and at each time step, whether to rely more on the general system state (from the shared layer) or its own specialized features.

For instance, during a sudden change in lithology, the gate for the ROP task might learn to increase the weight of its GeoNet expert, while during stable drilling, it might rely more on the shared layer's holistic representation. The resulting fused state vector (I_k) serves as the input to the corresponding upper-level tower network for the final prediction.

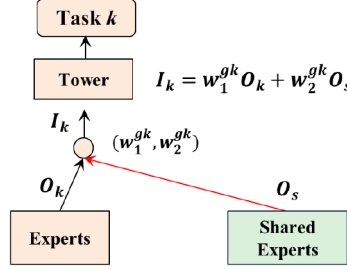


Figure 9—Information sharing between the shared layer and task-specific layers.

$$I_k = w_1^{gk} O_k + w_2^{gk} O_s \quad (1)$$

Where O_k and O_s represent the information extracted by the k -th task's lower-level experts and the shared layer, respectively. w_1^{gk} and w_2^{gk} are the weights output by the k -th task's gating unit, and I_k is the resulting weighted state vector.

Collaborative Training with Loss Function Balancing

In MTL, the total loss is a weighted sum of the individual task losses, as shown in Equation (2). Balancing these losses is crucial to prevent any single task from dominating the training process.

$$L_{MTL} = \sum_i w_i L_i \quad (2)$$

where L_{MTL} is the overall loss, w_i is the weight for the i -th task's loss function, and L_i is the loss value for that task.

We explored four different loss balancing strategies: (1) Fixed Weighting (FW): Assigns equal, fixed weights to the loss of each task (i.e., $w_i=1$). This is the simplest method but may not be optimal when tasks have different loss scales or learning difficulties. (2) Adaptive Weighting (AW): Dynamically adjusts the weights in each training epoch. As shown in Equation (3), the weight for each task is inversely proportional to its loss value. The AW method assumes that differences in the magnitude of task losses can affect convergence. Therefore, it assigns relatively higher weights to tasks with lower loss values, increasing their influence on the total loss, while assigning smaller weights to tasks with higher loss values to prevent them from unbalancing the training process. (3) Uncertainty Weighting (UW) (Kendall et al., 2018): Treats the weight for each task loss as a trainable parameter, learning the optimal weights based on the task's homoscedastic uncertainty. Conceptually, this method allows the model to learn to down-weight tasks that are inherently noisier or more difficult, preventing them from destabilizing the training of other tasks. A regularization term is included to prevent the weights from converging to zero. (4) Dynamic Weight Averaging (DWA) (Liu et al., 2019): Adjusts weights based on the rate of change of each task's loss. This strategy focuses the model's capacity on the tasks that are "struggling" the most. Tasks that are learning more slowly (i.e., their loss decreases at a slower rate) are given higher weights to accelerate their training.

$$w_i = \frac{1}{L_i} \quad (3)$$

Dataset and Preprocessing

The dataset used in this study was sourced from field drilling data from 14 wells in a block of the Tarim Oilfield, China. It includes 24 types of measurement and logging parameters, as detailed in Table 1. After a data preprocessing pipeline—which included data cleaning (Tunkiel et al., 2022) such as linear interpolation, mean imputation, outlier removal using the boxplot method, and denoising with a Savitzky-Golay filter, followed by data alignment and normalization—the final dataset comprised approximately 40,000 data points for modeling. Model hyperparameters, such as sequence length and learning rate, were systematically tuned using Bayesian optimization (Wu et al., 2019) to ensure optimal performance. The performance of the models was evaluated using three standard metrics: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2). These metrics assess the model's accuracy, relative error, and ability to explain the variance in the data, respectively.

Table 1—Parameters used for modeling in this study.

Parameter Name (Unit)		
Rate of Penetration (m/h)	Bottom Hole Pressure (MPa)	Formation Pressure (MPa)
Depth (m)	Time (s)	Weight on Bit (kN)
Rotary Speed (RPM)	Flow Rate (m ³ /s)	Torque (kN·m)
Hook Load (kN)	Standpipe Pressure (MPa)	Mud Pit Volume (m ³)
Mud Density (kg/m ³)	Mud Viscosity (Pa·s)	Inlet Temperature (°C)
Outlet Temperature (°C)	Outlet Flow (%)	Inlet Conductivity (mS/cm)
Outlet Conductivity (mS/cm)	Sonic Transit Time	Gamma Ray
Spontaneous Potential	Mud Resistivity	True Resistivity

Results and Discussion

Evaluation of Loss Balancing Strategies

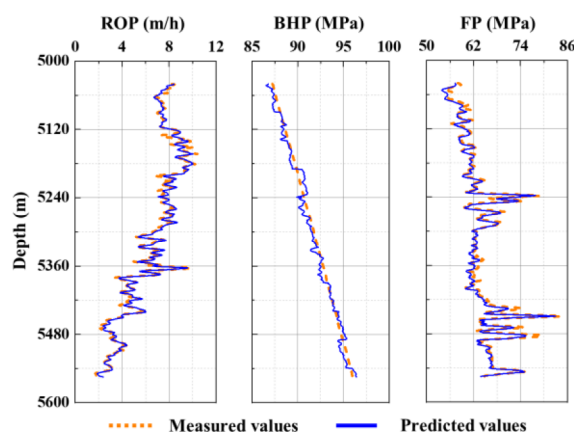
We first evaluated the performance of the four loss balancing strategies. The experimental results (Table 2) show that the choice of strategy significantly impacts the overall model performance. The Fixed Weighting (FW) strategy performed the worst, with an MAPE for ROP prediction approaching 10%. This is likely because the larger loss values of the BHP and FP tasks dominated the gradient updates, thereby suppressing the effective learning of the ROP task. In contrast, the three dynamic balancing strategies—AW, UW, and DWA—all achieved excellent performance, with ROP prediction MAPE values below 6% and R^2 values above 0.96. The purpose of these strategies is to enable different tasks to learn at a similar pace. AW, UW, and DWA adjust loss values through their unique methods to ensure gradient magnitude consistency, whereas FW's simple equal weighting can lead to insufficient updates for certain tasks, as was the case for ROP prediction here. Among all strategies, the Adaptive Weighting (AW) strategy achieved the best overall performance across the three prediction tasks. It not only delivered the highest accuracy for ROP prediction but also showed robust performance for BHP and FP. Its core advantage lies in its ability to dynamically ensure that different tasks learn at comparable rates. Therefore, all subsequent analyses in this paper are based on the HMTL model trained with the AW strategy.

Table 2—Comparison of prediction results under four loss balancing strategies.

Strategy	Task	MAE	MAPE	R ²
FW	ROP	0.4188 m/h	9.16%	0.9236
	BHP	0.5055 MPa	0.55%	0.9355
	FP	0.8038 MPa	1.26%	0.9568
AW	ROP	0.2100 m/h	3.74%	0.9853
	BHP	0.3356 MPa	0.36%	0.9760
	FP	0.9420 MPa	1.45%	0.9287
UW	ROP	0.2507 m/h	4.90%	0.9775
	BHP	0.4705 MPa	0.52%	0.9361
	FP	1.1843 MPa	1.83%	0.9025
DWA	ROP	0.3088 m/h	5.61%	0.9640
	BHP	0.3935 MPa	0.43%	0.9581
	FP	1.3497 MPa	2.00%	0.6225

HMTL Model Performance Analysis

The HMTL model trained with the AW strategy demonstrated excellent predictive capability and generalization performance. As seen in the prediction curves (Fig. 10), the model's predictions align closely with the true values. It accurately captures the dynamic trends of ROP, BHP, and FP under different formation and operating conditions. Even in sections with sharp fluctuations (such as ROP around 5360 m, and FP around 5240 m and 5480 m), the model maintains good tracking performance. The error distribution plots (Fig. 11) show that the prediction errors for each task follow a normal distribution centered around zero, indicating that the model's predictions are unbiased. The comprehensive performance metrics (AW strategy row in Table 2) further confirm the model's high accuracy.

**Figure 10—Prediction performance of the MTL model with the AW strategy.**

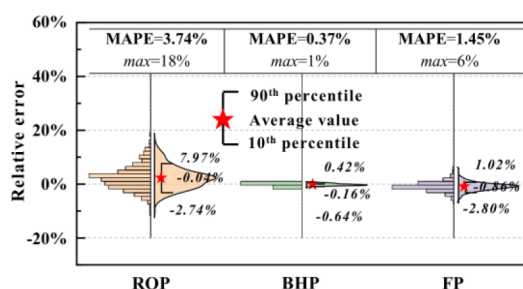


Figure 11—Relative prediction error distribution of the MTL model with the AW strategy.

Comparative Analysis: HMTL vs. STL

To validate the superiority of the HMTL framework, we conducted a comprehensive comparison of its performance against the independent STL models. As shown in the results in Table 2 and the visualization in Fig. 12, the HMTL model with AW consistently outperformed the STL models on all three tasks. The improvement in ROP prediction was particularly significant, with the MAE and MAPE decreasing by approximately 50%. The prediction errors for BHP and FP were also reduced by about 30% and 28%, respectively. This result strongly demonstrates that by learning inter-task correlations through the shared layer, the HMTL framework can acquire more robust and comprehensive feature representations than the independent models, thus overcoming the performance bottleneck of single-task learning. In addition to its accuracy advantage, the HMTL framework also excelled in computational efficiency. By integrating the training of three separate models into one, the average model update (training) time was reduced from 1301 seconds to 939 seconds, saving approximately 28% in computational time. In summary, the experimental results conclusively show that the HMTL method surpasses the traditional STL approach in both predictive accuracy and computational efficiency.

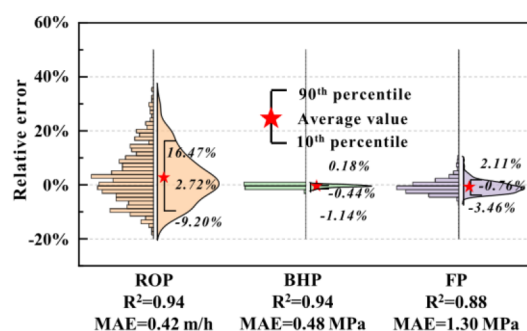


Figure 12—Prediction performance of the Single-Task Learning (STL) models.

Conclusion

To address the limitations of traditional single-task learning, which ignores the intrinsic correlations among drilling parameters, this paper successfully designed and validated a novel Heterogeneous Multi-Task Learning (HMTL) framework. This framework achieves efficient, collaborative prediction of three key drilling parameters: ROP, BHP, and FP, by intelligently integrating specialized, heterogeneous models into a single, unified architecture. The proposed HMTL architecture, with its unique design featuring "task-specific expert networks, a shared expert layer, and adaptive gating," effectively solves the challenge of achieving knowledge sharing among disparate models while preserving their individual, domain-specific strengths. Our experiments confirmed that dynamic loss balancing is crucial for effective collaborative training, with the Adaptive Weighting (AW) strategy yielding the best performance. The resulting HMTL model demonstrated significant accuracy improvements across all tasks compared to independent STL models, with error reductions of up to 50%, while also reducing total training time by approximately 28%.

This research provides a powerful and efficient paradigm for intelligent drilling, demonstrating that multi-task learning is a highly effective tool for solving complex, multi-variable engineering problems. The framework's scalability lays a solid foundation for future globally optimized predictive systems that can integrate additional drilling parameters, paving the way for safer, more efficient, and more autonomous drilling operations.

Acknowledgments

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Nomenclature

AW	= Adaptive Weighting
BHP	= Bottom Hole Pressure
DWA	= Dynamic Weight Averaging
FCNN	= Fully Connected Neural Network
FP	= Formation Pressure
FW	= Fixed Weighting
GRU	= Gated Recurrent Unit
HMTL	= Heterogeneous Multi-Task Learning
MAE	= Mean Absolute Error
MAPE	= Mean Absolute Percentage Error
MoE	= Mixture-of-Experts
MTL	= Multi-Task Learning
PIHNN	= Physics-Informed Hybrid Neural Network
R^2	= Coefficient of Determination
ROP	= Rate of Penetration
STL	= Single-Task Learning
UW	= Uncertainty Weighting

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